

## A Technical Introduction to Universal approximation theorem

$$f(x) = \sum_{i=1}^N v_i \sigma(w_i^T x + b_i)$$

### Section 1

**Bounded depth and bounded width case** The first result on approximation capabilities of neural networks with bounded number of layers, each containing a limited number of artificial neurons was obtained by Maierov and Pinkus. Their remarkable result revealed that such networks can be universal approximators and for achieving this property two hidden layers are enough.

### Section 2

In the field of machine learning, the universal approximation theorems (UATs) state that neural networks with a certain structure can, in principle, approximate any continuous function to any desired degree of accuracy. These theorems provide a mathematical justification for using neural networks, assuring researchers that a sufficiently large or deep network can model the complex, non-linear relationships often found in real-world data. The best-known version of the theorem applies to feedforward networks with a single hidden layer. It states that if the layer's activation function is non-polynomial (which is true for common choices like the sigmoid function or ReLU), then the network can act as a "universal approximator." Universality is achieved by increasing the number of neurons in the hidden layer, making the network "wider." Other versions of the theorem show that universality can also be achieved by keeping the network's width fixed but increasing its number of layers, making it "deeper." These are existence theorems. They guarantee that a network with the right structure exists, but they do not provide a method for finding the network's parameters (training it), nor do they specify exactly how large the network must be for a given function. Finding a suitable network remains a practical challenge that is typically addressed with optimization algorithms like backpropagation.

### Section 3

**Arbitrary-depth case** The "dual" versions of the theorem consider networks of bounded width and arbitrary depth. A variant of the universal approximation theorem was proved for the arbitrary depth case by Zhou Lu et al. in 2017. They showed that networks of width  $n + 4$  with ReLU activation functions can approximate any Lebesgue-integrable function on  $n$ -dimensional input space with respect to  $L^1$  distance if network depth is allowed to grow. It was also shown that if the width was less than or

equal to  $n$ , this general expressive power to approximate any Lebesgue integrable function was lost. In the same paper it was shown that ReLU networks with width  $n + 1$  were sufficient to approximate any continuous function of  $n$ -dimensional input variables. The following refinement, specifies the optimal minimum width for which such an approximation is possible and is due to.

### Section 4

**Variants** Variants include discontinuous activation functions, noncompact domains, certifiable networks, random neural networks, and alternative network architectures and topologies. The universal approximation property of width-bounded networks has been studied as a dual of classical universal approximation results on depth-bounded networks. For input dimension  $d_x$  and output dimension  $d_y$  the minimum width required for the universal approximation of the  $L_p$  functions is exactly  $\max\{d_x + 1, d_y\}$  (for a ReLU network). More generally this also holds if both ReLU and a threshold activation function are used. Universal function approximation on graphs (or rather on graph isomorphism classes) by popular graph convolutional neural networks (GCNs or GNNs) can be made as discriminative as the Weisfeiler–Leman graph isomorphism test. In 2020, a universal approximation theorem result was established by Br  el-Gabrielsson, showing that graph representation with certain injective properties is sufficient for universal function approximation on bounded graphs and restricted universal function approximation on unbounded graphs, with an accompanying  $O(|V| \cdot |E|)$ -runtime method that performed at state of the art on a collection of benchmarks (where  $V$  and  $E$  are the sets of nodes and edges of the graph respectively). There are also a variety of results between non-Euclidean spaces and other commonly used architectures and, more generally, algorithmically generated sets of functions, such as the convolutional neural network (CNN) architecture, radial basis functions, or neural networks with specific properties.

### Section 5

**Arbitrary-width case** A universal approximation theorem formally states that a family of neural network functions is a dense set within a larger space of functions they are intended to approximate. In more direct terms, for any function  $f$  from a given function space, there exists a sequence of neural networks  $\phi_1, \phi_2, \dots$  from the family, such that  $\phi_n \rightarrow f$  according to some criterion.

### Section 6

Reservoir computing and quantum reservoir computing In reservoir computing a sparse recurrent neural network with fixed weights equipped of fading memory and echo state property is followed by a trainable output layer. Its universality has been demonstrated separately for what concerns networks of rate neurons and spiking neurons, respectively. In 2024, the framework has been generalized and extended to quantum reservoirs where the reservoir is based on qubits defined over Hilbert spaces.

### Section 7

Setup Artificial neural networks are combinations of multiple simple mathematical functions that implement more complicated functions from (typically) real-valued vectors to real-valued vectors. The spaces of multivariate functions that can be implemented by a network are determined by the structure of the network, the set of simple functions, and its multiplicative parameters. A great deal of theoretical work has gone into characterizing these function spaces. Most universal approximation theorems are in one of two classes. The first quantifies the approximation capabilities of neural networks with an arbitrary number of artificial neurons ("arbitrary width" case) and the second focuses on the case with an arbitrary number of hidden layers, each containing a limited number of artificial neurons ("arbitrary depth" case). In addition to these two classes, there are also universal approximation theorems for neural networks with bounded number of hidden layers and a limited number of neurons in each layer ("bounded depth and bounded width" case).

### Section 8

Quantitative bounds The question of minimal possible width for universality was first studied in 2021, Park et al obtained the minimum width required for the universal approximation of  $L_p$  functions using feed-forward neural networks with ReLU as activation functions. Similar results that can be directly applied to residual neural networks were also obtained in the same year by Paulo Tabuada and Bahman Ghahserifard using control-theoretic arguments. In 2023, Cai obtained the optimal minimum width bound for the universal approximation. For the arbitrary depth case, Leonie Papon and Anastasis Kratsios derived explicit depth estimates depending on the regularity of the target function and of the activation function.